

Abstract No.

Identification of a Kinase-Autophagy Signaling Hub as a Therapeutic Vulnerability in Pediatric Glioma

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Pediatric glioma remains a devastating malignancy characterized by a profound lack of effective therapeutic interventions. This study identifies a specific stress-responsive kinase as a pivotal regulator of autophagy in pediatric glioma cells. Autophagy is a cellular recycling process essential for maintaining homeostasis and ensuring tumor survival under nutrient, energy, and amino acid deprivation. Through SILAC-MS/MS, co-immunoprecipitation, and colocalization analyses, we demonstrated that this kinase directly interacts with the key autophagy proteins, an association that is significantly enriched during autophagy-inducing stress. Functional validation reveals that the kinase is essential for pathway activation, as its knockout significantly impairs autophagy, whereas its overexpression is sufficient to initiate the process under basal conditions and amplify it under stress. Furthermore, our proteomics-based mapping of the kinase interactome and its phosphorylation targets uncovered several novel partners. By establishing this kinase as a central modulator of autophagic flux, our findings provide critical insights into basic cellular mechanisms and highlight the targeting of this kinase-autophagy axis as a promising therapeutic strategy for the treatment of pediatric glioma.



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